

A Systematic Review on a Multi-tenant Database Management System in Cloud Computing

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Abstract— As Cloud Computing (CC) continues to revolutionize, the demand for efficient and secure data management solutions has never been greater. Multi-tenant database Management Systems (MTDBMS) in the cloud offer the scalability and cost-effectiveness the organizations demand, but introduce complex security challenges that must be addressed to ensure the discretion, reliability, and accessibility of sensitive data. Hence, this survey scrutinizes 25 literature works focussed on MTDBMS in the context of CC and brings to light the various limitations of the existing methodologies. The primary objective of this survey is to investigate the various facets of MTDBMS, emphasizing the unique requirements and challenges posed by multi-tenancy (MTY) in the cloud. Further, the key architectural components of such systems are explored and highlight the importance of securing tenant data. Furthermore, this survey delves into the various security threats and vulnerabilities that MTDBMS may face in the cloud environment. This survey proves to be a valuable resource for researchers seeking comprehensive insights into the complexities of MTDBMS in CC. It offers a thorough assessment of the methodologies used, existing challenges, and future work to provide valuable insights into potential solutions, all contributing to the advancement of secure and efficient MTDBMS in cloud environments.

Keywords— Cloud computing; Multi-Tenant; Securing tenant data; Database Management; Cloud environments

I. INTRODUCTION

CC stands out as one of the most prominent and widely embraced technologies today, representing a prime example of Computing as a Utility. Within this paradigm, users embrace the "pay-as-you-go" type, where users pay for the computing, storage, and application resources the entities consume [1],[2]. The allure of CC further extends to its exceptional elasticity, allowing users to seamlessly scale resources up or down as needed, rendering it an attractive choice for organizations of all sizes [3]. The economic appeal of CC cannot be overstated. It incentivizes adoption by offering a cost-effective alternative, especially for enterprises seeking the delicate balance between elasticity and infrastructure costs. Many businesses grappling with the decision to construct their personal IT infrastructure to leverage Cloud communications often discover that the latter provides a more favorable cost-to-elasticity ratio [2],[4]. At its core, CC is characterized as "a system where data center resources are shared through virtualization technology, delivering elastic, on-demand, and instantaneous services to users, all while charging them based on their actual usage, much like a utility bill"[1]. It

encompasses the fundamental principles of Elasticity, complemented by On-Demand provisioning, Broad network accessibility, enormity, and Virtualization, which collectively define the quintessential attributes of the CC Model.

CC, as a service model, empowers users to efficiently access information system resources via the internet or network, covering their needs like data storage, processor power, and software from external sources using a pay-per-use approach. This model offers numerous business advantages, including cost reduction in hardware, software, and maintenance, as well as delivering high performance and flexibility [5],[6]. To maximize these benefits, services provided to clients should adhere to a "multi-tenant" architectural structure, optimizing resource sharing among users and reducing costs per customer [7],[8]. However, CC's benefits also introduce significant challenges, with security standing out as a paramount concern. Information security's primary objective is safeguarding data and computer systems against unauthorized entry, misuse, exposure, and other potential risks. In accordance with findings from the Cloud Security Alliance (CSA), the adoption of cloud computing is accompanied by seven significant threats. These include the improper use of cloud resources, vulnerabilities in application programming interfaces (APIs), the presence of malicious insiders, shared technology vulnerabilities, potential data leakage, instances of account and service hijacking, and an ever-evolving risk landscape. Addressing these challenges is vital for organizations to harness CC's advantages while safeguarding against potential risks.

Security concerns in CC are determined by the features of shared data and MTY. The cloud environment introduces an increased threat of data compromise due to the growing number of parties involved, which in turn leads to a higher number of access points [9]. Moreover, placing data control in the hands of cloud providers can heighten the potential for data vulnerability, as external services might circumvent the security measures implemented by users, encompassing personal, logical, and physical safeguards. Numerous apprehensions emerge surrounding MTY and the persistence of data. MTY, in the context of CC, involves resource sharing where any resource object can be reused within the cloud infrastructure. The management of reusable objects becomes critical since the model pose a substantial vulnerability and can potentially breach confidentiality by facilitating data leakage. This data leakage can be attributed to the lack of hardware separation in CC; While there exists a reasonable

degree of segregation at the application and virtual layers, the level of hardware separation remains insufficient [10]. Furthermore, there is a potential risk to confidentiality due to the possibility of data remanence when resource objects are reused. This situation could enable a customer to solicit storage space from a cloud provider and then perform scans for sensitive data belonging to other users [9],[11]. One of the most significant challenges in researching security within CC revolves around the transaction between cost and security, a critical factor influencing the adoption of cloud services.

The primary objective of this research is to gain a deeper understanding of Multi-tenant Database Management Systems in CC with,

- **Comprehensive Analysis:** To conduct an in-depth examination of the architecture, security mechanisms, and deployment models of Secure Multi-tenant Database Management Systems in the context of CC.
- **Highlight Emerging Solutions:** To showcase emerging technologies and strategies designed to enhance the security of MTDM in the cloud, with a focus on innovative approaches and advanced threat detection systems. The rest of this paper is structured as follows:

The structure of this paper is outlined as follows: Section I provides a brief introduction to the survey. Section II offers an Overview of CC, a comprehensive review of the existing literature is shown in Section III, while in Section IV, the analysis and discussion of the survey findings is elaborated. Sections V and VI highlight the identified research gaps, dataset used and potential avenues for future work, and finally, Section 6, brings the paper to a conclusion.

II. BACKGROUND AND OVERVIEW

CC technology offers efficient resource utilization for end-users. It embodies several key characteristics, including manageability, scalability, and availability. Moreover, CC boasts numerous merits like cost-effectiveness, on-claim provisioning, ease of use, widespread accessibility, MTY support, flexibility, and robust stability. CC primarily delivers services through three models: Platform as a Service (PaaS), Infrastructure as a Service (IaaS), and Software as a Service (SaaS). It also encompasses a range of deployment options, including public cloud, private cloud, hybrid cloud, community cloud, and virtual private cloud, as detailed in Figure 1.

In general, MTY serves as the foundational technology in CC, enabling the efficient and secure sharing of IT resources. An appropriate analogy is to envision a multitenant (MT) cloud environment like an apartment building, where numerous tenants efficiently share the building's common infrastructure while maintaining distinct walls and doors to ensure privacy from one another. Similarly, cloud technology leverages MTY to securely distribute IT resources among multiple applications and diverse tenants, including businesses and organizations, all benefiting from the advantages of cloud services while maintaining their data and operational privacy.

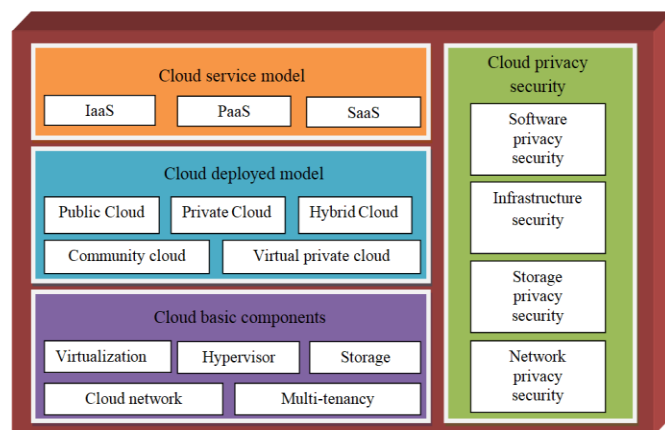


Fig. 1. Cloud Computing Framework

III. TAXONOMY OF MULTITENANCY ON CLOUD COMPUTING

The taxonomy diagram of MTY on CC is shown in Figure 2, where a detailed description is shown in the following sections.

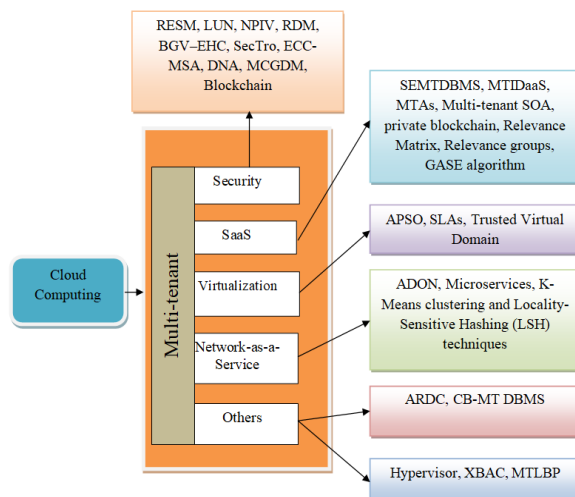


Fig. 2. Taxonomy of Multitenancy on Cloud Computing

A. Multitenant security services

In [10] Arun Kumar Yadav et al introduce SA2-MCD, an innovative architecture planned to improve the security and efficiency of multi-tenant cloud databases (SA2-MCD) within a public cloud setting. This architecture offers a reliable private window within the broader MT environment. SA2-MCD plays a pivotal role in optimizing transactions, ultimately delivering both robust software and hardware-level security. To bolster software-level security, the model employs Database as a Service (DBaaS) components. Future enhancements aim to amplify the utilization of resources by up to 95% while concurrently minimizing power consumption and reducing economic costs by up to 20%.

In [12] Pooja Dhiman et al introduced a novel model that seamlessly integrates multi-factor authentication and authorization processes. This innovative approach enhances security for tenants in a hybrid manner, combining the

strengths of two encryption methodologies. The proposed model demonstrated remarkable efficiency in safeguarding data from ciphertext attacks while achieving a high success rate in data transmission among multi-tenants. Future plans involve the inclusion of additional test cases, specifically aimed at securing data transmitted among multi-tenant servers. These plans also entail the incorporation of eight-byte data, to fortify data against outside attacks during transfer.

To ensure the secure transmission of data, protect privacy, and maintain data integrity, Babangida Albaba Abubakar conducted comprehensive research as outlined in [13]. This research focused on analyzing the security requirements of Cloud-Enabled Small Cell (CESC) systems using the Secure Tropos (SecTro) methodology. The research extensively investigated the security challenges inherent in CESC deployments and proposed effective mitigation strategies. The research was motivated by the Small cell coordination for MTY and Edge services network architecture. The analysis performed in the research emphasized the importance of securing the entire CESC system. Consequently, the research recommends the adoption of an MTY virtual environment within the SESAME architecture to bolster security measures and enhance the overall resilience of CESC systems.

K. Pawan, and A. K. Bhatt.[14] utilized the elliptic curve cryptography (ECC) which enabled authorized individuals to securely access private data while effectively safeguarding against various potential attacks. To enhance the security of data encryption, the research combines a nature-inspired optimization technique, specifically the Moth Search Algorithm (MSA), with ECC. This fusion aims to identify the most suitable and optimal elliptic curve value. Additionally, the proposed encryption and decryption methodology integrates Deoxyribonucleic Acid (DNA). The incorporation of DNA with ECC offers additional security while minimizing computational overhead. Future work will prioritize the development of an efficient strategy for dynamic resource allocation and a load-balancing mechanism tailored for multi-tenant cloud storage environments.

In a recent research conducted by Suyel Namasudra et al [15] a pioneering data encryption approach rooted in DNA-based computing revolves around the generation of a 1024-bit secret key, amalgamating DNA computing principles, user attributes, and the user's Media Access Control (MAC) address. By employing decimal encoding rules, and DNA base pairing, this technique crafts a robust secret key that fortifies the system against a spectrum of security threats. Future work in the form of a comprehensive security analysis, including mathematical proofs, and opportunities exist for enhancing the authentication procedures within the CC environment, marking an exciting avenue for future research and development.

In [7] Sarah Maroc et al introduce a novel approach to evaluate multi-tenant cloud services. Unlike traditional methods that focus on individual users, their framework centers around group evaluations, ensuring that the diverse requirements of all tenants are met. The process begins by evaluating the performance of various cloud service providers and comparing them to the specific needs of the tenant groups.

To establish the significance of each tenant's input objectively, the model employed a fuzzy TOPSIS technique, taking into account factors such as risk attitudes and contributions to the collective decision. Future investigations will expand upon the work by developing an integrated framework that accommodates diverse consensus models and factors, such as tenant SLA classes and subscription types (e.g., enterprise or standard users). This expanded framework aims to provide more comprehensive guidance and enhancement to the overall decision-making process in multi-tenant cloud service selection.

In [16] Matevs Pusticek delves into the security considerations surrounding decentralized applications, with a specific focus on the security of smart contracts. Decentralized ledgers bring about unique demands in the realm of smart contract design, development, deployment, and maintenance. The paper offers a concise overview of their approach to crafting secure smart contracts, shedding light on critical vulnerabilities associated with smart contracts and the developer tools designed to fortify their security. Notably, the model provides, a secure, and modular multi-contract platform capable of upgrades. Even when partly implemented on a public blockchain network, their solution manages to maintain transaction costs at a reasonable level, avoiding substantial increases in contrast to single-contract solutions. This stage is positioned to meet the demands of various services and can be seamlessly scaled to accommodate the requirements of modern 5G environments.

B. Software as a Service (SaaS) environment

In [9] G. B. Pallavi introduces a novel and secure MTDBMS (SEMTDBMS) for the CC. SEMTDBMS begins by conducting a thorough analysis of the security necessities of tenants, proposing a novel security weight metric to aid in the selection of tenant workers. Additionally, this work introduces an innovative workload scheduler designed to efficiently distribute workloads among tenants. The overall outcomes of this research reveal noteworthy improvements in reduced latency, and enhanced throughput when compared to existing models. Future research endeavors will involve assessing SEMTDBMS under diverse workloads and benchmarks, with a continued focus on enhancing both security measures and the scheduling mechanism.

In [3], Mohamed Yassin et al introduced a novel concept known as the MT Intrusion Detection as a Service (MTIDaaS), designed to facilitate seamless integration within Software as a Service (SaaS) environments. The MTIDaaS framework has undergone rigorous integration and testing within a real-world public cloud setting, demonstrating its practicality and effectiveness. This innovative framework offers SaaS capabilities, benefiting both the SaaS provider and individual tenants, all while maintaining a high degree of adaptability, portability, and cost-efficiency. By providing a versatile and portable Intrusion Detection System (IDS), MTIDaaS ensures that safety measures are accessible and shared among all subscribed tenants, thereby minimizing the resource overhead and cost for all parties involved.

D. Nikola et al.[17] underscored the importance of MTY as a fundamental requirement for cost-effective Internet of

Things (IoT) solutions on cloud platforms discussing various approaches to implementing MTY and thoroughly investigating the associated security considerations and data isolation measures in SaaS. Furthermore, it introduces a novel solution geared towards enhancing data layer scalability to meet performance requirements effectively. Additionally, the paper advocates for a service-oriented architecture (SOA) as a means to deliver a secure MT web service with adaptable data layers. This proposed framework comprises three distinct services, ultimately creating a highly secure and robust multi-tenant solution.

Aditi Sharma [18] introduces the concept of utilizing a private blockchain to implement an MT-based storage system. The primary aim is to create a highly scalable arrangement while ensuring the security and integrity of tenants' data. This proposed system harnesses the advantages of blockchain technology to achieve both scalability and robust data isolation among different tenants within the system. Notably, their research demonstrates that each tenant's data remains entirely independent of others and is exclusively accessible to authorized users. Furthermore, the system exhibits resilience; even in cases where certain tenants are deactivated, the system continues to function seamlessly, thanks to effective communication among the remaining operational nodes.

In [19], L. Xiaona, and J. Zhao introduced an innovative approach to address the challenge of managing data partitions in Software as a Service (SaaS) environments. The model identified the significant issue in existing cloud data management systems, where data partitions often lead to numerous distributed transactions. To mitigate this problem, the three-layered MT data partition model introduces concepts of multiple nodes for multi-tenant data using shared storage resources, thus improving the efficiency and scalability of SaaS data management.

M. R. Sumalatha, and M. Parthiban. [20] explored the unique challenges associated with MTY testing in SaaS environments, offering an analysis of the dynamic configuration differences between SaaS testing and traditional application testing. It introduces novel concepts like performance testing and the concept of a "fittest" function for each tenant. In this context, the paper utilizes the GASE (Genetic Algorithm and Symbolic Execution) algorithm as the foundation for the fitness function, which represents a promising approach to address the intricate testing demands of multi-tenant SaaS applications.

In their work, Antonio Rico et al. [21] introduce an innovative data model extension designed to improve upon traditional Multi-Tenant Architectures (MTAs). This extension, referred to as "multi-target," enables MT applications to efficiently handle and provide numerous functionalities within a single MT environment. This pioneering approach introduces exciting prospects for future research, as the multifunctional environment it creates brings forth novel challenges. For example, achieving a balance in resource management becomes crucial; in scenarios like MT^2 , where two high-computational-demand functionalities coexist within the same instance, there is a need to address potential

performance bottlenecks and ensure optimal system performance.

C. Virtualization

In [22] Vijay Varadharajan and Udaya Tupakula introduced an innovative architectural framework, trusted virtual domains. This architecture is designed to meticulously consider the security policies governing the tenant domains, and the individual virtual machines within these tenant domains. Notably, this proposed framework goes beyond just policy considerations; it also encompasses robust mechanisms aimed at bolstering the security assurance of communications occurring among virtual machines that reside in separate domains.

Designing a secure access mechanism to effectively manage MTY within virtualization is imperative. Virtualization encounters significant challenges related to resource allocation and establishing trust among its users. Praveen Pachauri et al., in their work [23], address the crucial issue of handling data migration among virtual machines. Furthermore, evaluating the service level agreements offered by the service provider is of paramount importance. These agreements should comprehensively outline all the promised services, including detailed associated charges. As cloud applications continue to expand, it becomes increasingly vital to adopt advanced technologies that can fully harness the benefits of virtualization while ensuring trust, security, and convenient access for all stakeholders.

The concept of MTY encompasses both public and private cloud models, spanning all three layers of the CC system. In this paper, an innovative approach known as Adaptive Particle Swarm Optimization is introduced to tackle the MTY aspect, facilitating the efficient utilization of resources within a cloud storage network. This research is primarily focused on addressing resource allocation challenges within a cloud environment, with the ultimate goal of enhancing performance for both cloud consumers and providers, thereby benefiting businesses. Future directions for this research include the development of methods for detecting virtual machine attacks using reusable objects, with a particular emphasis on mitigating potential data breaches and vulnerabilities within multi-zone environments [2].

D. Network as a Service:

F. Andrei et al.[24] provides a detailed exploration of MT components and microservices, offering a practical technique for proactively identifying possible vulnerabilities through static code analysis. This methodology is not only useful for pinpointing potential defects but can also serve as a means to ensure that both multi-tenant components and microservices adhere to the necessary security standards and legal requirements associated with cloud applications, bolstering their overall robustness and compliance.

In [6] Yanchun et al propose two distinct methods for recommending services in MT service-based systems (SBSs), one applicable during system construction (build-time) and the other during system operation (runtime). These approaches leverage advanced techniques, specifically K-Means clustering and Locality-Sensitive Hashing (LSH). Their

primary objective is to efficiently identify suitable services within such systems. The clustering-based service recommendation method has demonstrated notable effectiveness and efficiency, particularly in scenarios where meeting tenants' varying Quality of Service (QoS) requirements proves challenging and unevenly distributed. As part of their future research endeavors, the model intend to explore alternative clustering algorithms to address the limitations associated with K-Means.

In [13] Ronny Bazan Antequera et al introduce an innovative concept called "Application-driven Overlay Network-as-a-Service" (ADON), which effectively addresses the hybrid cloud needs of numerous applications. ADON empowers users to define specific requirements for their applications, which are then seamlessly translated into the essential network and computing resources. As part of their forthcoming research agenda, the approach intends to collaborate on additional data-intensive application use cases, encompassing a wide spectrum of computing and networking requirements.

E. Other Multitenant based approaches

CC offers numerous advantages, yet its widespread adoption is hindered primarily by concerns surrounding privacy. This is attributed to the fact that when individuals or organizations transition to commercial public cloud services provided by separate administrative entities, the approach relinquishes direct control over the systems responsible for managing their data and applications. This shared infrastructure, made possible by hardware virtualization, enables multiple users to simultaneously operate distinct application instances, thereby enhancing resource efficiency. However, this unique MTY characteristic also introduces novel security and privacy challenges in user interactions. To address these issues, a proposed model known as the "authorization-role distribution center" issues access tickets for requested services, contingent on users accepting the cloud's service level agreements (SLA), with the majority of such centers utilizing symmetric key encryption [25]. In recent research [8], a cluster-based multi-tenant database management system (CB-MT DBMS) was introduced, which proactively addresses SLA violations by making migration and replication decisions ahead of time. To enhance the efficiency of this system, a dynamic proactive MT database migration and replication algorithm (MTDB-MR) was proposed to minimize conflicts and inconsistencies among tenant migrations and replications. A promising avenue for future research is the extension of the MTDB-MR algorithm to incorporate diverse prediction models for forecasting [11]. In related work, a unified access control model known as XBAC was presented, tailored for heterogeneous MTY architectures in the cloud environment. XBAC utilizes learning automata to manage access policies and requests, with plans to expand its capabilities to support flexible collaborations among tenants at various delegation levels. Another critical aspect of cloud security is highlighted in [1] which proposes a scalable and secure access control mechanism designed specifically for MTY cloud environments, aiming to thwart malicious tenant activities. Future endeavors in this area involve prototyping and

implementation within a real cloud environment. Lastly, [5] introduces an innovative load-balancing strategy called MT load balance (MTLBP), tailored to the service-oriented architecture of MTY in the cloud. This approach guarantees efficient task execution for end-client users operating within a multi-tenant environment, emphasizing the importance of optimizing cloud service delivery.

IV. ANALYSIS AND DISCUSSION

A. Analysis based on journals used

Table 1 presents a list of journals that were considered for the review of research papers related to Multi-tenant Database Management. This review encompassed journals from reputable publishers including Elsevier, IEEE, MDPI, Springer, Research Gate, and ScienceDirect. The analytical representation of this data can be visualized in Figure 3.

TABLE 1: JOURNALS ACCESSED

Journals	Accessible	Selected	Papers
IEEE	22	9	[23],[7],[8],[3],[16],[4],[22],[6],[14]
MDPI	18	1	[12]
Science Direct	7	3	[10],[15],[24]
Springer	19	6	[9],[2],[5],[17],[18],[19]
Research gate	27	5	[1],[11],[13],[25],[20]
Taylor & Francis	14	1	[21]

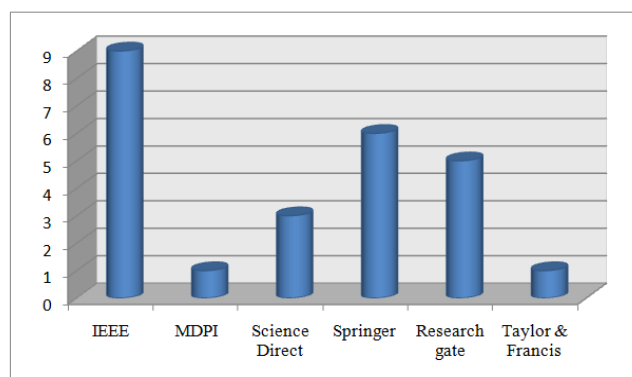


Fig.3. Analysis based on journals used

B. Analysis based on the year

To analyze Multi-tenant Database Management, the review gathered relevant journals based on a year-wise selection process, as illustrated in Figure 4, which gives information about the number of journals relevant to Multi-tenant Database Management are selected that were published in the respective years from 2012 to 2024.

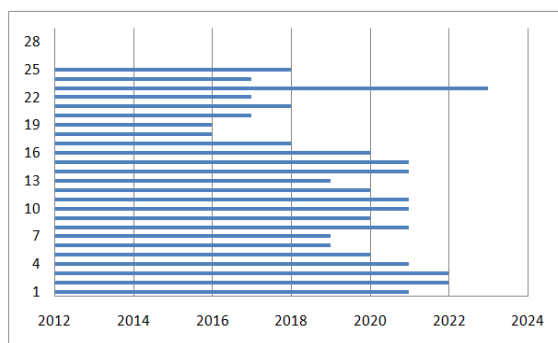


Fig. 4. Analysis based on year

V. DATASET USED

Some of the dataset used for the analysis of the research paper selected for this survey is described as follows

- TPC Benchmark™ H (TPC-H): TPC-H is an extensively used dataset for benchmarking decision support database systems. It consists of a large collection of business-related data across various domains, making it suitable for testing the performance and scalability of multi-tenant database systems. [8]
- TPC Benchmark™ DS (TPC-DS): TPC-DS is another decision-support benchmark dataset that simulates a data warehousing environment. It provides complex, real-world queries and data generation, making it valuable for assessing the capabilities of multi-tenant databases in handling analytical workloads [8].
- CloudSim 3.0.3 -CloudSim is an open-source toolkit that facilitates the simulation and modeling of CC environments and resource allocation strategies. As of my last knowledge update in September 2021, CloudSim 3.0.3 was one of the versions of CloudSim. It provides a Java-based simulation framework for researchers, developers, and educators interested in researching and experimenting with CC systems without the need for physical cloud infrastructure [10].
- OLTP, or Online Transaction Processing, benchmark is a performance testing methodology used to assess the speed and efficiency of database systems when handling a large number of short, interactive transactions [9].
- MySQL is an open-source relational database management system (RDBMS) that is widely used for managing and organizing structured data. It is known for its speed, reliability, and ease of use, making it a popular choice for a broad range of applications, from small websites to large-scale enterprise systems [12][19].
- The QWS Web service dataset, which is openly accessible, encompasses over 2500 real-world web

services, each characterized by nine dimensions of Quality of Service (QoS), including response time. To facilitate our analysis, a random partitioning approach was employed dividing the web services into five distinct service classes.

- Amazon Elastic Compute Cloud (Amazon EC2), often abbreviated as EC2, is a web service provided by Amazon Web Services (AWS). It offers resizable computing capacity in the cloud, allowing users to launch and manage virtual machines (known as instances) to run a wide range of applications [18].

VI. RESEARCH CHALLENGES AND FUTURE WORK

Challenges and Future Work in Secure Multi-Tenant Database Management Systems in CC are as follows:

- Balancing security with performance is an ongoing challenge. Future research should work on optimizing query performance, especially in scenarios involving encrypted data, without compromising security.
- In the future, efforts will be concentrated on devising an efficient dynamic resource allocation strategy and load balancing mechanism for multi-tenant cloud storage, with the primary goal of mitigating DDoS attacks.
- Future work encompassing a comprehensive security analysis, complete with mathematical proofs, presents opportunities to enhance authentication procedures within the CC environment, marking an intriguing pathway for future research and development
- Adhering to evolving data regulations requires continuous effort. Automation of compliance checks, real-time auditing, and blockchain-based auditing trails could be areas for further exploration.
- Advanced threat detection systems that leverage machine learning and anomaly detection techniques should be developed to proactively identify and mitigate security threats within multi-tenant databases.
- Effective encryption strategies, including homomorphic encryption and post-quantum cryptography, need to be integrated seamlessly while addressing key management complexities. Developing efficient and scalable key management solutions is essential.
- Efficient resource allocation mechanisms and transparent billing models are crucial. Research should focus on optimizing resource allocation for multi-tenant databases while ensuring fair billing practices.
- As organizations increasingly adopt multi-cloud and hybrid cloud strategies, future work should address secure multi-tenant database management across diverse cloud providers and environments.

- Enhancing user-friendliness of security measures, such as integrating secure defaults and user training, is essential to reduce human-related security risks.
- Research should explore techniques for auto-scaling multi-tenant databases to accommodate variable workloads efficiently while maintaining security.
- Secure communication between tenants and preventing cross-tenant interference are critical challenges. Future work should devise methods for secure inter-tenant communication and isolation.
- Ensuring robust data isolation between tenants remains a significant challenge. Future work should focus on developing advanced techniques like secure containers and data anonymization to enhance data privacy, particularly in shared infrastructure models.
- Addressing these challenges and advancing research in these areas will be instrumental in creating robust and secure multi-tenant database management systems in the ever-evolving landscape of CC, ensuring the confidentiality, integrity, and availability of tenant data.

VII. CONCLUSION

In conclusion, as the landscape of IT continues to be reshaped by the relentless evolution of CC, the need for robust and secure data management solutions has never been more pronounced. MTDBMS in the cloud holds the promise of scalability and cost-efficiency, but they also present formidable security challenges that demand our unwavering attention. This survey has provided a thorough exploration of the dynamic realm of MTDBMS within the context of CC. To enhance the security and efficiency of multi-tenant databases, future research should prioritize refining fine-grained access control mechanisms, optimizing query performance in encrypted data scenarios, automating compliance checks, developing advanced threat detection systems leveraging machine learning, integrating seamless encryption strategies, enhancing the user-friendliness of security measures, ensuring secure inter-tenant communication, and developing advanced data isolation techniques while adhering to evolving data regulations. As a valuable resource, this survey equips researchers, practitioners, and decision-makers with the knowledge needed to navigate the intricacies of secure MTDBMS in CC. Identifying the prevailing security concerns and suggesting potential solutions, paving the way for a future where multi-tenant database management in the cloud can be both secure and efficient, thereby realizing the full potential of this transformative technology.

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